



Practice assignment 10 - Not Graded



This assignment will not be graded and is only for practice.

1 point

Match the functions of two variables in Column A with their directions in which the directional derivatives at (0,0) is maximized given in Column B.

☐

(1) $\rightarrow$ (v)(1) $\rightarrow$ (v)

☐

(1) $\rightarrow$ (iii)(1) $\rightarrow$ (iii)

☐

(2) $\rightarrow$ (iv)(2) $\rightarrow$ (iv)

☐

(3) $\rightarrow$ (iv)(3) $\rightarrow$ (iv)

☐

(2) $\rightarrow$ (vi)(2) $\rightarrow$ (vi)

☐

(4) $\rightarrow$ (v)(4) $\rightarrow$ (v)

☐

(3) $\rightarrow$ (vi)(3) $\rightarrow$ (vi)

☐

(4) $\rightarrow$ (i)(4) $\rightarrow$ (i)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1) $\rightarrow$ (iii)(1) $\rightarrow$ (iii)

(2) $\rightarrow$ (iv)(2) $\rightarrow$ (iv)

(4) $\rightarrow$ (v)(4) $\rightarrow$ (v)

1 point

Which of the following statements is(are) correct?

☐

Maximum directional derivative at (0,0) for the function $f(x, y) = |x| + |y|$ is 1.

☐

The graph of the linear approximation to a function $f(x, y)$ at a given point P is the tangent plane to $f(x, y)$ at the point P .

☐

A tangent plane to a function $f(x, y)$ exists at every point.

☐

All directional derivatives of a function $f(x, y)$ at a point P are 0 if the tangent plane is parallel to the XY -plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The graph of the linear approximation to a function $f(x, y)$ at a given point P is the tangent



plane to $f(x, y)$ at the point P .

All directional derivatives of a function $f(x, y)$ at a point P are 0 if the tangent plane is parallel to the XY -plane.

1 point

Which of the following is(are) critical points of the scalar valued multivariable function

$$f(x, y) = 6x^2 - 2x^3 + 3y^2 - 6xy \quad f(x, y) = 6x^2 - 2x^3 + 3y^2 - 6xy?$$

☐ (1,1)

☐ (0,0)

☐ (1,-1)

☐ No critical points.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1,1)

(0,0)

Consider two functions f_1, f_2 defined as

$$f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2) \quad f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2)$$

and

$$f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2) \quad f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2).$$

If $L_{f_1}(x, y, z)$ and $L_{f_2}(x, y, z)$ are the linear approximation of $f_1(x, y, z)$ and $f_2(x, y, z)$ at point $(1, -1, 1)$, then find the value of

$$85(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z)) \quad 85(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z)).$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 30.0

1 point

$z = f(x, y) = -\frac{7}{6}x^2 + \frac{1}{3}y^2 + \frac{11}{6}z = f(x, y) = 6 - 7x^2 + 31y^2 + 611$ represents a surface. Let (x_0, y_0, z_0) be the point where the tangent plane to the given surface is parallel to the plane $7x - 8y + 3z + 2 = 0$. Find the value of $11x_0 + 5y_0 + 4z_0$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 55

1 point

Find the number of points where the tangent plane to the function

$f(x, y) = x^3 + y^3 + 3x^2 - 9x - 12y + 20$ at the points is parallel to the XY -plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

- Suppose $ax + by + cz = d_1$ is a tangent plane to the function $f(x, y) = x^3 + 3y^3 + 9xy$ at $(-1, 2)$ and $ax + by + cz = d_2$ is a tangent plane to the same function $f(x, y)$ at $(\alpha, 1)$.

Use the above information to answer questions 7-8:

1 point

Which of the following options is(are) true?

☐

$$\alpha = -2$$

☐

$$\alpha = 2$$

☐

There are only two possible candidates for α .

☐

$$|d_2 - d_1| = 12$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\alpha = 2$$

$$|d_2 - d_1| = 12$$

Find the value of $\frac{2a+b+c}{d_1+d_2}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

- Let $L(x, y)$ be the linear approximation to $f(x, y) = xe^y + \ln(x + y)$ at point $(1, 0)$.

Use the above information to answer questions 9-10:

Suppose $L(x, y) = ax + by + c$, where $a, b, c \in \mathbb{R}$. Find the value of $a + b - c$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 5

1 point

Find the approximate value of the function $f(x, y)$ at $(1.1, 0.1)$ using the linear approximation function $L(x, y)$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1.4

1 point

[Check Answers](#)

Your score is: 0/10